Tables

Table 1. Demographic and Audiometric Characteristics of the NHANES Cohort

Characteristic	Training (n=15,654 ears)	Test (n=3,914 ears)
Participants (n)	9,421	3,519
Age (years), mean (SD)	43.9 (14.4)	44.0 (14.4)
Age (years), median (IQR)	44 (31–56)	44 (31–56)
Female sex	50.9%	51.3%
Mean PTA-4 (dB HL)	13.3 (11.4)	13.3 (11.6)
Normal	88.0%	88.6%
Mild	8.8%	8.1%
Moderate	2.2%	2.3%
Moderately Severe	0.6%	0.7%
Severe	0.3%	0.3%
Profound	0.1%	0.0%

Table 2. Classification Performance on Held-Out Test Set

Metric	FAI (Fuzzy)	PTA-4 (WHO)	XGBoost	Random Forest
Weighted vs. Reference	0.93	1.00*	0.98	0.98
Overall Agreement	95.8%	100%*	98.8%	98.8%
Borderline Agreement (±5 dB)	84.9%	100%*	93.5%	93.8%
Clear-case Agreement	98.4%	100%*	100%	100%
Spearman vs. PTA-4	0.80	1.00	1.00	1.00
MAE vs. PTA-4 (dB)	6.1	0.0*	0.2	0.5

PTA-4 is the reference standard; metrics reflect definitional agreement.

Table 3. Configuration Distribution in the NHANES Cohort (Exploratory)

Configuration	Prevalence (%)	Slope Range
Flat	56.7%	−5 to 12 dB
Gently Sloping	18.0%	12–28 dB
Steeply Sloping	8.8%	28–45 dB
Precipitous	3.5%	>45 dB
Rising	13.0%	<−5 dB

Table 4. Fuzzy vs. Crisp Classification at the Normal–Mild Boundary

PTA-4 (dB)	WHO Grade	FAI Score	FAI Grade	Key Memberships
24	Normal	11.3	Normal	Normal =0.33, Mild =0.00
26	Mild	25.7	Mild	Normal =0.16, Mild =0.40
27	Mild	28.9	Mild	Normal =0.07, Mild =1.00
28	Mild	30.0	Mild	Normal =0.00, Mild =1.00
30	Mild	30.0	Mild	Normal =0.00, Mild =1.00

Table 5. Optimised trapezoidal membership function parameters (a, b, c, d) for the six hearing loss severity categories.

Category	a	b	c	d
Normal	0.0	0.0	16.4	27.8
Mild	25.8	26.3	36.4	43.5
Moderate	41.5	42.0	54.3	58.5
Moderately Severe	56.5	57.0	67.9	73.5
Severe	71.5	72.0	83.7	93.0
Profound	91.0	91.5	104.6	120.0

Table 6. Comparator models evaluated on the held-out test set.

Model	Description	Target
PTA-4 (WHO)	Pure-tone average of 0.5, 1, 2, 4 kHz; classified using WHO crisp thresholds (25/40/55/70/90 dB)	Severity grade
XGBoost	Gradient-boosted regression trees (learning rate 0.1, max depth 6, 100 estimators); input = 7 frequency thresholds; output = continuous PTA-4 estimate	PTA-4 (continuous)
Random Forest	1,000 regression trees (max depth 10, min samples split 5); input = 7 frequency thresholds; output = continuous PTA-4 estimate	PTA-4 (continuous)

Table 7. Table (context: Classification Accuracy)

Method	vs. Reference	Borderline Agreement (±5 dB)	Clear-case Agreement
FAI (fuzzy)	0.93	84.9%	98.4%
PTA-4 (WHO)	1.00*	100%*	100%*
XGBoost	0.98	93.5%	100%
Random Forest	0.98	93.8%	100%

PTA-4 is the reference standard; its agreement is definitional. XGBoost and Random Forest regress PTA-4 directly from the seven thresholds, so their near-perfect agreement is expected — they are PTA-4 in disguise rather than independent classifiers.

Table 8. Classification at the normal/mild boundary: crisp labels versus graded fuzzy output across the 24–30 dB HL range.

PTA-4	Crisp Label	Fuzzy Label	FAI Score	Normal	Mild
24 dB	Normal	Normal	11.3	0.33	0.00
26 dB	Mild	Mild	25.7	0.16	0.40

27 dB	Mild	Mild	28.9	0.07	1.00
28 dB	Mild	Mild	30.0	0.00	1.00
30 dB	Mild	Mild	30.0	0.00	1.00

Table 9. Summary of the four clinical case studies (A–D).

Case	PTA-4	WHO Grade	FAI	Configuration	Key Finding
A	27.5 dB	Mild	29.4	Precipitous (shape rules)	Borderline membership (Normal 0.04, Mild 1.00)
B	23.0 dB	Normal	21.5	Precipitous (shape rules)	25 dB notch at 4 kHz masked by PTA
C	33.8 dB	Mild	30.0	Sloping	Frequency-resolved severity preserved
D	53.8/28.8 dB	Mod/Mild	54.5 composite	Precipitous	Asymmetry upgrade; inter-aural modulation